CLICK HERE CLICK HERE

APUL

NEET FLIX

AIM FREE EDUCATION TO ALL

NEET CHANNEL

JOIN OUR TELEGRAM CHANNELS HERE

TEST PAPERS <<< CLICK

TOPPER'S NOTES <<< CLICK

BEST FROM KOTA LECTURES <<< CLICK

QUIZES <<< CLICK

REVISION FLASH CARDS &

MANY MORE <<< CLICK

BIOLOGY

1) **Assertion :** Mosses along with lichens are the first organisms to colonize rocks.
Reason : They decompose rocks by secreting acid substances.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

2) **Assertion :** *Gelidium* and *Gracilaria* are member of rhodophyceae.
Reason : They are autotrophic, marine organism and they have predominance of the red pigment r-phycoerythrin in their body.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

3) Consider the following w.r.t pteridophytes
Statement I: The diploid sporophyte is represented by dominant, independent, photosynthetic, vascular plant body.
Statement II: It alternates with multicellular, autotrophic, independent but short-lived haploid gametophyte.

(1) Both are correct
(2) Both are incorrect
(3) Statement I is correct and statement II is incorrect
(4) Statement I is incorrect and statement II is correct

4) **Assertion :** Algae increase the level of dissolved oxygen in their immediate environment.
Reason : Half of total CO_2 fixation on earth is carried out through photosynthesis by algae.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

5) **Assertion :** *Cycas* is an example of dioecious plant.
Reason : In *Cycas* male cone and megasparophylls are produced on different plant.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

6) **Assertion (A) :** Events precursor to seed habit was initially observed in pteridophytes.
Reason (R) : Female gametophyte retained on the parent sporophyte for variable period in pteridophytes

(1) Both (A) and (R) are true and (R) is correct explanation of the (A).

(2) Both (A) and (R) is true but (R) is not correct explanation of the (A).

(3) (A) is true but (R) is false

(4) Both (A) and (R) is false

7) Kelps are related with :-

(1) Chlorophyceae

(2) Phaeophyceae

(3) Rhodophyceae

(4) Cyanophyceae

8) Which of the following have multicellular sporophyte which is not free living but attached to photosynthetic gametophyte and derives nourishment from it ?

(1) Pteridophytes

(2) Gymnosperms

(3) Bryophytes

(4) Thallophytes

9) The aquatic fern which is used as a biofertiliser is :-

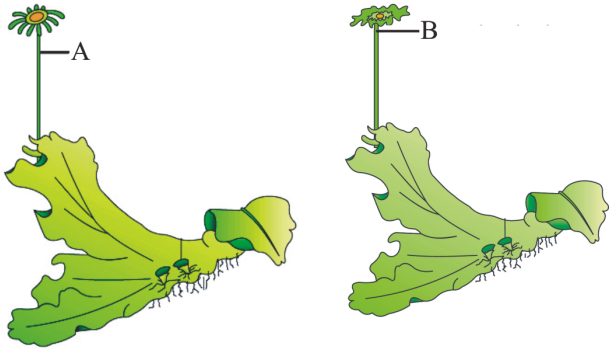
(1) *Adiantum*

(2) *Salvinia*

(3) *Azolla*

(4) *Dryopteris*

10) Recognise the figure and find out the correct matching :-



(1) A = Archegonia ; B = Antheridia

(2) A = Archegoniophore; B = Antheridiophore

(3) A = Antheridiophore; B = Archegoniophore

(4) A = Antheridia ; B = Archegonia

11) As we go from species to kingdom in a taxonomic hierarchy, the number of common characters will

(1) decrease

(2) increase

(3) remain same

(4) increase or decrease

12) ICBN is :-

(1) International Code of Biological Naming

(2) International Code of Botanical Nomenclature

(3) International Class of Biological Nomenclature

(4) International Classification of Biological Nomenclature

13) _____ is the branch of biology dealing with identification, nomenclature and classification of organisms

(1) Morphology

(2) Anatomy

(3) Ecology

(4) Taxonomy

14) How many of the following are taxa ?
Dogs, Cats, Mammals, Wheat, Rice, Animals, Plants, wood, leaf, table, Jointed legs.

(1) 12

(2) 7

(3) 9

(4) 11

15) Match the Column-I and Column-II and choose right options.

	Column-I		**Column-II**
(I)	Class	(P)	Category includes related class
(II)	Genus	(Q)	Category includes related orders
(III)	Phylum	(R)	Category includes related species
(IV)	Kingdom	(S)	Category includes related divisions

	(I)	(II)	(III)	(IV)
(1)	Q	R	S	P

(2)	P	Q	R	S
(3)	Q	R	P	S
(4)	S	R	Q	P

(1) 1

(2) 2

(3) 3

(4) 4

16) According to the rules of binomial system of nomenclature which of the following is not correct ?

(1) First word in biological name represents the genus while the second component denotes the specific epithet

(2) Specific epithet starts with capital letter while the genus starts with a small letter

(3) Both the words in a biological name, when handwritten, are seperately underlined

(4) Name of the author appears after the specific epithet

17) Select **correctly** written scientific name of Mango which was first described by Carolus Linnaeus:

(1) *Mangifera indica* Car. Linn.

(2) *Mangifera indica* Linn.

(3) *Mangifera indica*

(4) *mangifera Indica*

18) **Statement-I** : Each genus may have one or more than one specific epithets representing different organism but having morphological similarities.
Statement-II : Panthera has specific epithet called tigris and solanaceae is a genus it includes nigrum and melongena.

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct and Statement-II is incorrect

(4) Statement-I is incorrect and Statement-II is correct

19) Which one of the following belongs to the family Muscidae ?

(1) Fire fly

(2) Grasshopper

(3) Cockroach

(4) House fly

20) Genera *Petunia* and *Datura* belongs to family-

(1) Solanaceae

(2) Fabaceae

(3) Liliaceae
(4) Cucurbitaceae

21)

The classification of Linnaeus was mainly based on :-

(1) Vascular tissue
(2) Androecium
(3) Sepals
(4) Overall morphology of plants

22) Equal weightage to both vegetative and reproductive characters is given in -

(1) Artificial system of classification
(2) Natural system of classification
(3) Numerical system of classification
(4) Both (1) & (3)

23)

Find the incorrect from the following statements.

(1) To know about diversity and relationship of organism is systematics
(2) The scope of systematics was later enlarged to include identification, nomenclature and classification of organisms
(3) Systematics takes into account evolutionary relationships among organisms
(4) The earliest classifications of organisms were based on evolutionary relationships in organisms

24) Which of the following are responsible for the production of methane in the gut of several ruminant animals ?

(1) Halophiles
(2) Bacillus
(3) *E.coli*
(4) Methanogens

25) Choose the incorrect statement of following :-

(1) Ribosomes are the site of protein synthesis.
(2) Pili do not play a role in motility.
(3) Inclusion bodies are bounded by membrane.
(4) Cell wall prevent the bacterium from bursting and collapsing.

26) Which option is not related with *mycoplasma* ?

(1) Smallest living cell

(2) Faculative anaerobic

(3) Lack cell membrane

(4) Pathogenic in plants and animals

27) Bacterial structure and behaviour are respectively.

(1) Simple, Simple

(2) Complex, Simple

(3) Simple, Complex

(4) Complex, Complex

28) Single protective unit of bacteria is made up of:-

(1) Cell wall and cell membrane

(2) Glycocalyx, cell wall and flagella

(3) Cell wall, cell membrane and pili

(4) Capsule, cell wall and cell membrane

29) Read the following point carefully.
(a) Nitrogen fixation ability
(b) Oxygenic photosynthesis
(c) Have **chlorophyll a** similar to green plants
These points are related with :

(1) Eubacteria

(2) Euglenoids

(3) Green algae

(4) Cyanobacteria

30) How many organisms are included in Monera in given list ?
(A) Slime moulds (B) Dinoflagellates
(C) Protozoans (D) *Pseudomonas*
(E) Archebacteria (F) Basidiomycetes

(1) 2

(2) 4

(3) 3

(4) 5

31) Choose the correct match :-

(1) Mesosomes - Cell respiration

(2) Plasmid - bear photosynthetic pigments

(3) Flagella - help in formation of conjugation tube

(4) Capsule - help in locomotion

32) Heterocysts are found in :-

(1) *Anabaena* and *Nostoc*

(2) *Mucor* and *Chlorella*

(3) *Euglena* and *Paramoecium*

(4) *Lichens* and *Agaricus*

33) Which of the following is incorrect about blue green algae ?

(1) They may be unicellular, colonial or filamentous

(2) Their colonies are generally surrounded with membranous sheath

(3) They often form blooms in polluted water bodies

(4) Some of the members can fix atmospheric nitrogen

34) Consider the following statements:-
(a) They grow in soil, on logs and tree stumps and in living plant bodies as parasites. (b) The mycelium is branched and septate and the asexual spores are generally not found but vegetative reproduction by fragmentation is common. (c) The sex organs are absent but plasmogamy is brought about by fusion of two vegetative or somatic cells of different strains or genotypes. Above statements are related to the member of which class :-

(1) Phycomycetes

(2) Ascomycetes

(3) Basidiomycetes

(4) Deuteromycetes

35) What is/are demerit(s) of two kingdom system of classification?

(1) It did not distinguish prokaryotes and eukaryotes

(2) It did not distinguish unicellular and multicellular organisms

(3) It did not separate photosynthetic and non-photosynthetic organisms

(4) All of the above

36) **Assertion:** Deuteromycetes are referred as imperfect fungi
Reason: Sexual stage is not known in these fungi.

(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both the assertion and reason are false.

37) **Assertion-(A) :-** According to Whittaker classification *Chlorella* and *Amoeba* both are included in kingdom Protista
Reason-(R) :- *Chlorella* and *Amoeba* both are unicellular eukaryotes and have cellulosic cell wall

(1) A is true while R is false

(2) A and R both are true & R is not correct explanation of A

(3) A is false while R is true

(4) A and R both are true and R is correct explanation of A

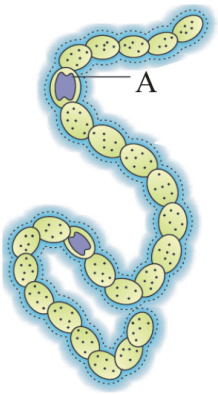
38) **Assertion :** Archaebacteria are able to survive in harsh conditions.
Reason : Presence of peptidoglycan in cell wall help archaebacteria to survive in extreme conditions.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.



39) Identify 'A' in the figure given below and to which process it is related ?

(1) Holdfast - Nitrogen fixation

(2) Heterocyst - Nitrogen fixation

(3) Heterocyst - Production of methane

(4) Pyrenoids - Storage bodies



40) Which of the following is correct for given diagram?

(1) Asexual spore are endogenously formed in it

(2) Sexual spores are exogenously formed in it

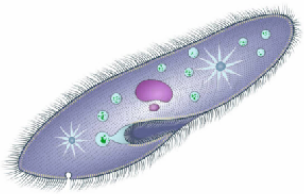
(3) It is a sac fungi

(4) Zygospores are formed in it

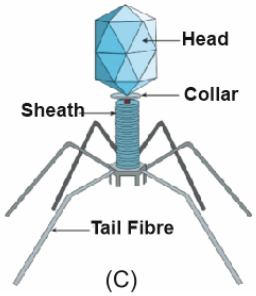
41) Identify figures A, B, C and D :



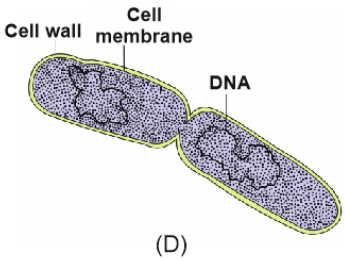
(A)



(B)



(C)



(D)

(1) *A = Agaricus, B = Paramoecium, C = Bacteriophage, D - Anabaena*

(2) *A= Aspergillus, B = Euglena, C = TMV, D = Dividing bacteria*

(3) *A = Aspergillus, B = Paramoecium, C = Bacteriophage, D = Dividing bacteria*

(4) *A = Mucor, B = Euglena, C = TMV, D = Nostoc*

42) Consider the following four statements (A–D) and select the option which includes all the incorrect option only.
(A) M.W. Beijernik demonstrated that the extract of the infected plants of tobacco could cause infection.
(B) W.M. Stanley showed that viruses could be crystallised.
(C) Lichen are the symbiotic relationship between the root of higher plants and fungi
(D) The RNA of the viroid is of high molecular weight.

(1) Only C and D

(2) Only B, C and D

(3) Only A, B and D

(4) A, B, C and D

43) Consider the following statements :-
I. In this group, the *Plasmodium* differentiates and forms fruiting bodies, bearing spores at their tips.
II. Spores possess true walls.
III. The spores are dispersed by air currents.
IV. The spores are extremely resistant and survive for many years even under adverse conditions.
The above statements are assigned to :-

(1) Euglenoids

(2) Slime moulds

(3) Dinoflagellates

(4) Chrysophytes

44) Given below are two statements :
Statement-I : In Ascomycetes, asexual spore are conidia, produced exogenously on the special

mycelium called conidiophore.
Statement-II : In Ascomycetes, sexual spore are called ascospores which are produced endogenously in sac like asci.
In the light of the above statement,
Choose the most appropriate answer from the option given below :

(1) Both statement-I and statement-II are correct.
(2) Both statement-I and statement-II are incorrect.
(3) Statement-I is correct but statement-II is incorrect.
(4) Statement-I is incorrect but statement-II is correct.

45) **Statement-1:** Heterocysts are found in nitrogen fixing BGA.
Statement-2: Some of BGA can fix atmospheric nitrogen in specialised cells called heterocysts.

(1) Both statement 1 and statement 2 are correct
(2) Statement 1 is incorrect but statement 2 is correct
(3) Statement 1 is correct but statement 2 is incorrect
(4) Both statement 1 and statement 2 are incorrect

46) What is common among *Porphyra, Agaricus, Chlorella, Spirulina* ?

(1) They all are photosynthetic
(2) They have eukaryotic cell organization
(3) They all are single celled
(4) They all used as food or food supplements

47) Read the following term carefully.
Cilia, Flagella, Zygote, Photosynthesis, Pellicle, Peptidoglycan wall, Nitrogen fixation ability, Multicellular, heterocyst, decomposer.
How many terms from these are not related with kingdom protista:-

(1) 4
(2) 5
(3) 7
(4) 6

48) Identify the correct match from the column I, II, and III :-

Column-I		**Column-II**		**Column-III**	
1	Chlorophyceae	a	Conidia	i	Fucus
2	Phaeophyceae	b	Pyrenoid	ii	Somatogamy
3	Ascomycetes	c	Clamp connection	iii	Equal length flagella
4	Basidiomycetes	d	Fucoxanthin	iv	Penicillium

(1) 3-b-ii, 4-c-i, 2-a-iii, 1-d-iv

(2) 3-d-i, 3-b-iii, 4-c-iv, 1-a-ii

(3) 1-b-iii, 2-d-i, 3-a-iv, 4-c-ii

(4) 1-a-ii, 2-c-iii, 3-b-iv, 4-b-i

49) Which one of the following statements is **wrong**?

(1) Agar-Agar is obtained from *Gelidium* and *Gracilaria*.

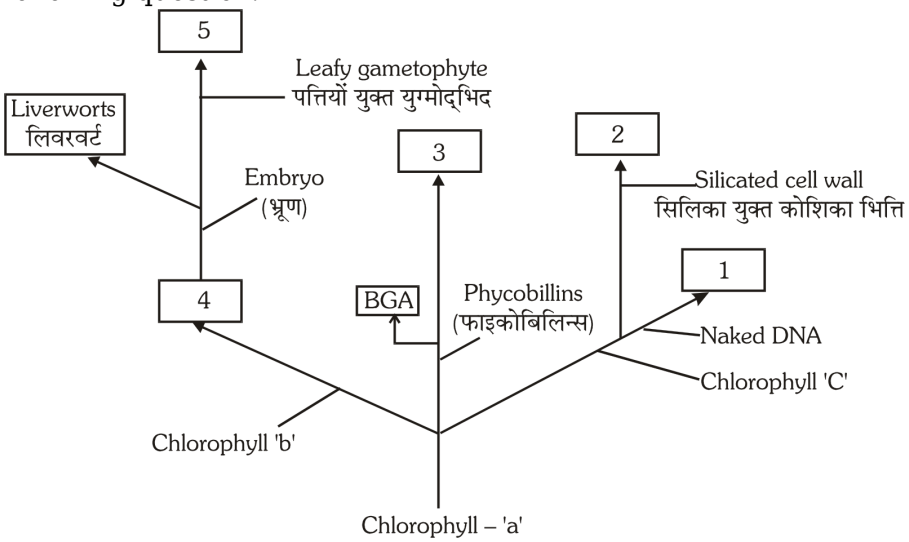
(2) *Chlorella* and *spirulina* are used as space food.

(3) Mannitol is stored food in phaeophyceae.

(4) Algin and carragen are products of fungi.

50)

A cladogram showing evolution of plants given below. Study it carefully and give the answer of following question.



Number 1 representing the which group of organism ?

(1) Moss

(2) Dinoflagellates

(3) Diatoms

(4) Brown algae

51) Which one is a **wrong** statement ?

(1) Brown algae have cellulosic cell wall covered by algin

(2) Green algae may store food in the form of oil droplet

(3) Red Algae have biflagellate zoospores

(4) Chlorophycean members are grass green in colour

52) Choose the **correct** option of members of Rhodophyceae :-

(1) *Ectocarpus*, *Porphyra*, *Fucus*

(2) *Polysiphonia, Porphyra, Gelidium*

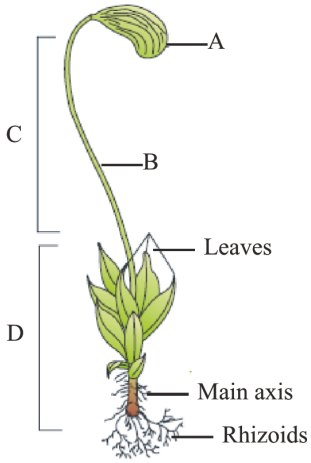
(3) *Dictyota, Chara, Gelidium*

(4) *Gelidium, Ectocarpus, Polysiphonia*

53) Which one of the following statements is wrong?

(1) Algin and carrageen are products of algae
(2) Agar-agar is obtained from *Gelidium* and *Gracilaria*
(3) *Chlorella* and *Spirullina* are used space travellers as a food
(4) Mannitol is stored food in Rhodophyceae

54) In given figure A,B,C and D represent : -

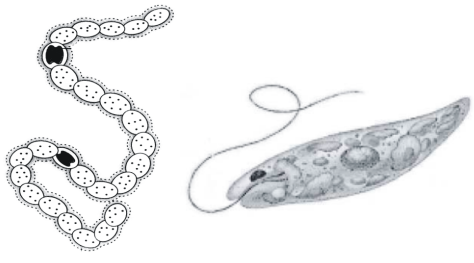


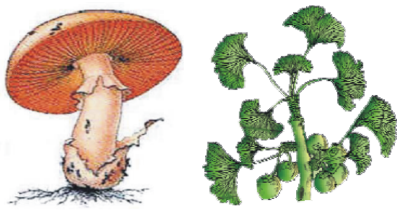
(1) A-Capsule, B-seta, C-Gametophyte, D-Sporophyte
(2) A-Capsule, B-Seta, C-Sporophyte, D-Gametophyte
(3) A-foot, B-Seta, C-Gametophyte, D-Sporophyte
(4) A-foot, B-Capsule, C-Gametophyte, D-Sporophyte

55) In which of the following motile cells are not formed?

(1) Zygomycetes and oomycetes
(2) Zygomycetes and phycomycetes
(3) Chlorophyceae and phycomcetes
(4) *Mucor, Rhizopus and Spirogyra*

56)





Identify the A, B, C and D

(1) A - *Equisetum*, B - *Paramoecium*,
C - *Funaria*, D - *Sphagnum*

(2) A - *Euglena*, B - *Agaricus*,
C - *Ginkgo*, D - *Nostoc*

(3) A - *Agaricus*, B - *Ginkgo*,
C - *Euglena*, D - *Equisetum*

(4) A - *Nostoc*, B - *Euglena*,
C - *Agaricus*, D - *Ginkgo*

57) A protonema is :-

(1) A structure in pteridophytes formed before the thallus develops

(2) A sporophytic free living structure formed in pteridophytes

(3) A creeping, green, filamentous and gametophytic structure produced in bryophytes.

(4) A primitive structure formed after fertilization in pteridophytes

58) Which of the following class of algae rarely found in fresh water?

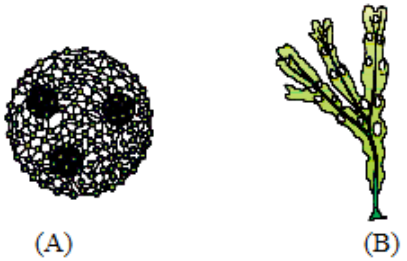
(1) Chlorophyceae

(2) Phaeophyceae

(3) Cyanophyceae

(4) Both (1) and (2)

59) The four sketches (A,B, C and D) given below represent four different organisms. Which one of these is correctly identified in the options given, along with its correct stored food and class ?



		Name of Organism	**Stored Food**	**Class**
(1)	(D)	*Nostoc*	Glycogen	Myxophyceae
(2)	(A)	*Volvox*	Floridean starch	Chlorophyceae
(3)	(B)	*Laminaria*	Starch	Phaeophyceae

(4)	(C)	*Polysiphonia*	Floridean starch	Rhodophyceae

(1) 1

(2) 2

(3) 3

(4) 4

60) **Assertion :** Bryophytes are also called amphibians of the plant kingdom.
Reason : Bryophyte can live in soil but are dependent on water for sexual reproduction.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

ANSWER KEYS

BIOLOGY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**1**	**1**	**1**	**1**	**1**	**1**	**2**	**3**	**3**	**2**	**1**	**2**	**4**	**2**	**3**	**2**	**2**	**3**	**4**	**1**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**2**	**4**	**4**	**4**	**3**	**3**	**3**	**4**	**4**	**1**	**1**	**1**	**2**	**3**	**4**	**1**	**1**	**2**	**2**	**3**
Q.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	**3**	**1**	**2**	**1**	**1**	**4**	**1**	**3**	**4**	**2**	**3**	**2**	**4**	**2**	**4**	**4**	**3**	**2**	**4**	**1**

SOLUTIONS

BIOLOGY

1) NCERT Page No. # 34

2) **Explain Question :** The question asks for the relationship between two statements: an Assertion and a Reason
Concept : This question is based on Rhodophyceae
Explanation :
Assertion : Gelidium and Gracilaria are member of rhodophyceae- True
Gelidium and Gracilaria are indeed classified under Rhodophyceae (Red Algae).
Reason : They are autotrophic, marine organism and they have predominance of the red pigment r-phycoerythrin in their body.- True
The members of Rhodophyceae are commonly called red algae because of the predominance of the red pigment, r-phycoerythrin, in their body. Most of the red algae are marine with greater concentrations found in warmer areas
The characteristics mentioned in the Reason (autotrophic, marine, r-phycoerythrin) are typical features of Rhodophyceae, which is why Gelidium and Gracilaria are classified under this group.
Final answer :1

3)

Question Explanation: Explaining the life cycle of pteridophytes.

Concept: Alternation of Generation

Solution: Both statements are correct. Pteridophytes have a life cycle where the dominant phase is the diploid, independent sporophyte. This alternates with a haploid gametophyte that is also multicellular, autotrophic, and free-living, but short-lived.

Final Answer: Option 1: Both are correct

4) **Explain Question :** The question presents an Assertion and a Reason and asks for a correct analysis of their relationship
Concept : This question is based on Algae, General Characters
Solution :
Assertion: "Algae increase the level of dissolved oxygen in their immediate environment."
This is True. Through photosynthesis, algae release oxygen into water, raising dissolved O_2 levels.
Reason: "Half of total CO_2 fixation on Earth is carried out through photosynthesis by algae."
This is also True. Algae, especially microscopic phytoplankton, perform about 50% of global CO_2 fixation.

Does the Reason explain the Assertion?
Yes. The CO_2 fixation through photosynthesis is directly linked to O_2 release.
Final answer : 1

5)

Explain Question : The question asks us to evaluate an Assertion and a Reason related to the plant Cycas.
Concept : This question is based on lower gymnosperms.
Explanation :
Assertion: "Cycas is an example of dioecious plant." - True.
Dioecious plants have separate male and female reproductive structures on different individual plants.
Reason: "In Cycas male cone and megasporophylls are produced on different plant." - True.
In Cycas, the male cones (bearing microsporophylls) are found on one plant, and the megasporophylls (bearing ovules) are found on a separate plant, which is the characteristic of a dioecious plant.
Relationship between Assertion and Reason:
The reason directly explains why Cycas is considered a dioecious plant by detailing the separation of male and female reproductive structures on different plants.
Final answer : 1

6)

NCERT Pg. # 38
Event Precursor to seed habit was initially observed in pteridophytes because some pteridophytes are heterosporous (microspore & megaspore). In heterosporous pteridophytes such as seleginella, marselia, azolla, and salvinia female gametophyte retain on the parent female sporophyte for variable period so seed habit nature evolved from pteridophytes but they do not produce seeds.

7) **Explain Question :**The question asks to which class of algae kelps belong
Concept : This question is based on Phaeophyceae
Explanation : The brown algae (phaeophyceae) are mostly marine and include forms such as the giant kelps, which may measure up to 100 metres in length.
Final answer : 2

8)

Explain Question : This question asks which plant group has a multicellular sporophyte that is not free-living but is attached to and derives nourishment from a photosynthetic gametophyte.
Concept : This question is based on Bryophyta, General Characters And Life Cycle
Explanation : In bryophytes sporophyte is not free-living but attached to the photosynthetic gametophyte and derives nourishment from it
Final answer :3

9)

Explain Question : The question asks for the name of a specific type of aquatic fern that is utilized as a biofertilizer.
Concept : This question is based on Pteropsida
Explanation : "Azolla, a water fern, has Anabaena (a nitrogen-fixing cyanobacterium) in its leaves and is used as a biofertiliser, especially in paddy fields."
Final answer : 3

10)

Explain Question : The question asks to correctly match the labels A and B.
Concept : This question is based on liverworts.
Explanation : A represents the Archegoniophore, which is the stalked structure bearing the female reproductive organs (archegonia).
B represents the Antheridiophore, which is the stalked structure bearing the male reproductive organs (antheridia).
Final answer : 2

11)

NCERT XI, Pg # 8

12)

NCERT XI, Pg # 4

13)

NCERT XI, Pg # 4

14) NCERT Page No. # 7 Para - 1.2
Dogs, Cats, Mammals, Wheat, Rice, Animals, Plants, Mangifera indica, Triticum aestivum.

15) NCERT Page No. # 9, 10

16)

Explain Question: Question is asking which is not correct about the rule of nomenclature.

Concept: Nomenclature

Solution:

1 First word in biological name represents the genus while the second component denotes the specific epithet--correct

2 Specific epithet starts with capital letter while the genus starts with a small letter-not correct

The first word denoting the genus starts with a capital letter while the specific epithet starts with a small letter.

3 Both the words in a biological name, when handwritten, are separately underlined--correct

4 Name of the author appears after the specific epithet--correct

Final answer: 2

17)

NCERT XI, Pg # 4

18) NCERT-XI, Pg#9 Para-1.3.1

19)

NCERT XI, Pg # 8

20)

NCERT XI Page # 9, Para 4, line 4,5,6

21)

NCERT XI, Pg # 5

22)

NCERT-XI, Pg. # 23, 24

23)

NCERT XI Pg # 3,4,5

24) **Explain Question:** Question asks about the organisms that produce methane in the gut of ruminant animals.
Concept: This question is based on Archaebacteria
Solution: Methanogens: Archaebacteria that produce methane in anaerobic conditions, found in rumen of cattle and buffalo.
Final answer: 4

25)

NCERT-XI, Pg. # 128, 129

26) **Explain Question:** Question asks about which statement is not related to Mycoplasma.
Concept: This question is based on Mycoplasma
Solution:
The incorrect statement is that Mycoplasma lack cell membrane.
Final answer: 3

27) NCERT XI Pg. # 19

28) **Explain Question:** Question asks about the single protective unit of bacteria.
Concept: This question is based on Structure of bacteria
Solution: The protective unit = capsule + cell wall + cell membrane.
Final answer: 4

29) **Question Explanation:** Identifying characteristics of an organism.
Concept: Cyanobacteria features
Solution: The listed features are all characteristic of **cyanobacteria**. They are a type of photosynthetic bacteria that can perform **oxygen-releasing photosynthesis** (oxygenic photosynthesis). They also possess **chlorophyll a**, a pigment similar to that found in green plants. Furthermore, many cyanobacteria, such as *Nostoc* and *Anabaena*, have specialized cells called heterocysts that are capable of **nitrogen fixation**.
Final Answer: Cyanobacteria

30)

Solution:

The Monera kingdom includes prokaryotic organisms, such as:

A. **Pseudomonas (a bacterium)**

B. **Archaebacteria (a type of archaea)**
Answer :- 1

31) **Option 1:** Correct. Mesosomes are invaginations of the plasma membrane in bacteria, involved in cellular processes like cell respiration.
Option 2: Incorrect. Plasmids are small, circular DNA molecules, not involved in bearing photosynthetic pigments.
Option 3: Incorrect. Flagella are involved in locomotion, not in the formation of conjugation tubes.
Option 4: Incorrect. Capsules protect bacteria and help in avoiding immune responses, not in locomotion.

32) **Explain Question:** Question asks in which organisms heterocysts are found.

Concept: Cyanobacteria

Solution: Heterocysts occur in Anabaena and Nostoc.

Final answer: 1

33)

Explain Question : The question asks to identify the incorrect statement among the given options regarding blue-green algae, also known as cyanobacteria.

Concept : This question is based on cyanobacteria.

Solution :
Their colonies are generally surrounded with membranous sheath: incorrect.
The colonies of blue-green algae are typically surrounded by a gelatinous (mucilaginous) sheath, not a membranous one.

Final Answer : option (2)

34)

NCERT XI, Pg. # 18

35)

Explain Question :
The question asks about the drawbacks of the two-kingdom classification system.

Concept :
This question is based on classification or taxonomy.

Solution :
The two-kingdom system failed to account for :
Cellular structure : No distinction between prokaryotes (like bacteria) and eukaryotes.
Body organization : No separation of unicellular organisms (like Amoeba) from multicellular ones.
Mode of nutrition : It grouped photosynthetic and non-photosynthetic organisms together.

Final Answer: All of the above
Option (4)

36)

Question Explanation: The question asks whether the assertion and reason about Deuteromycetes (imperfect fungi) are true and if the reason correctly explains the assertion.

Concept: Deuteromycetes

Solution: A-True Deuteromycetes are called imperfect fungi because their sexual stage is not known.

R- True This absence of sexual reproduction is the reason behind the term imperfect.

Final answer: Both the assertion and the reason are true and the reason is a correct explanation of the assertion.

37) **Explain Question:** Question is asking whether the assertion and reason about **Chlorella and Amoeba in Whittaker's classification** are true or false.
Concept: This quesion is based on Protista
Solution :
Assertion: Correct, since both Chlorella (unicellular alga) and Amoeba (unicellular protozoan) are placed in Protista.
Reason: Wrong, because Amoeba does not have a cell wall; only Chlorella has a cellulosic cell wall.
Assertion is true, but Reason is false.
Final answer : 1

38) NCERT, Pg. # 13

39)

Explain Question: Question asks to identify structure 'A' in the given figure and the process it is related to.

Concept: Cyanobacteria

Solution: 'A' represents a heterocyst, and its main function is nitrogen fixation.

Final answer: 2

40) **Explain Question:** Question is asking to identify the correct statement about the fungal structure shown in the diagram
Concept: This question is based on Ascomycetes
Solution : Given diagram is Aspergillus. It is a sac fungi because it belong from Ascomycetes.So correct answer is 3
Final answer : 3

41)

Generated by Allie

Problem Statement: The question asks to identify four biological figures labeled A, B, C, and D in the given images. Each figure represents a different organism or biological entity, and the task is to match each figure with its correct name.

Underlying Concept: This question involves basic biology and taxonomy concepts, focusing on identifying various microorganisms and viruses. Understanding the characteristics of fungi, protozoa, viruses, and bacteria is essential to differentiate and correctly identify each figure.

Tips and Tricks: Remember these key features: Aspergillus is a fungus with spore-bearing structures; Paramecium is a ciliated protozoan; Bacteriophage has a complex virus structure with head and tail fibers; Dividing bacteria show cell wall and DNA splitting.

Common Mistakes: Confusing similar-looking microorganisms such as Aspergillus and Mucor (both fungi), or mixing up protozoans like Euglena with ciliates like Paramecium. Also, confusing viruses (TMV) with bacteriophages.

Why Other Options Are Incorrect?: Choices involving TMV (Tobacco Mosaic Virus) are incorrect because the structure in (C) is a bacteriophage which has a distinct head and tail, unlike TMV. Identifying Euglena instead of Paramecium in (B) is wrong because Euglena is flagellated and not ciliated like Paramecium. Similarly, Nostoc and Anabaena are cyanobacteria, different from the dividing bacteria shown in (D).

42)

Explain Question : Pick which statements (A–D) are incorrect only from the four given.

Concept : This question is based on Virus, Viroids and Lichens.

Solution :
(A) True — Beijerinck showed infectious tobacco extract.
(B) True — W.M. Stanley crystallised tobacco mosaic virus.
(C) False — lichens are fungus + alga/cyanobacterium, not roots of higher plants.
(D) False — viroid RNA is low molecular weight (small circular RNA).

Final answer : option (1)

43)

Explain Question : Question is asking which group of organisms is described by the given statements about plasmodium, spore formation, and dispersal.

Concept : This question is based on slime moulds

Solution : The characteristics listed (plasmodium → fruiting bodies, resistant airborne spores) belong to slime moulds.

Final answer : option (2)

44) NCERT-XI, Pg. # 17, 18

45)

NCERT-XI, Pg. # 13

46)

NCERT Pg. No. 33

47)

NCERT XI, Pg # 14

48)

NCERT XI, Pg # 26

49) **Explain Question :** The question asks to identify the incorrect statement among four options related to the uses and characteristics of different organisms, particularly algae and fungi.
Concept : This question is based on Intra Of Algae And
Explanation : Algin and carrageenan are products of fungi. - Incorrect
Algin and carrageenan are not products of fungi; they are polysaccharides derived from brown algae (Phaeophyceae) and red algae (Rhodophyceae), respectively.
Final answer : 4

50)

NCERT XI, Pg # 15

51)

Explain Question : The question asks you to identify the incorrect statement among the four options provided.

Concept : This question is based on algal classes

Solution : Red algae (Rhodophyceae) are characterized by the absence of flagellated stages, meaning they do not have biflagellate zoospores.

Final Answer : option (3)

52)

Explain Question : The question asks to identify the group of algae that belongs to the class Rhodophyceae, commonly known as red algae
Concept : This question is based on Rhodophyceae - red algae
Explanation : “Members of Rhodophyceae (red algae) include Polysiphonia, Porphyra, and Gelidium.”
Final answer : 2

53) NCERT XI Pg.# 18,19,20

54)

Explain Question : The question has provided a diagram of a moss plant and its life cycle stages, labeled A, B, C, and D. The task is to correctly identify each labeled part and match it to one of the four given options.
Concept : This question is based on life cycle of Bryophytes - moss.
Explanation :
A: Capsule- This is the spore-producing part of the sporophyte, located at the apex.
B: Seta- This is the stalk that supports the capsule, connecting it to the gametophyte.
C: Sporophyte- This refers to the entire diploid structure, which includes the foot (embedded in the gametophyte), the seta, and the capsule.
D: Gametophyte- This represents the entire haploid plant body, which is the dominant stage in the moss life cycle and includes the leaves, main axis, and rhizoids.

Final answer : option (2)

55)

NCERT XI, Pg # 17

56)

NCERT XI, Pg.# 13 ,15, 17, 33

57) NCERT XI, Pg # 30

58)

NCERT XI, Pg # 26

59)

Explain Question : The question has provided an image with four sketches (A, B, C, and D) of different organisms and a table with four options. The task is to identify which option correctly matches the organism, its stored food, and its class.
Concept : This question is based on types of algae
Explanation :
Polysiphonia (C) is a red alga, belonging to the class Rhodophyceae, and stores food primarily as Floridean starch, which is characteristic of this group.
Final answer : 4

60) NCERT Pg. # 29